Title
A Statement of Work for the Procurement of a Hydrogen Peroxide Generation System

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Introduction
The State-of-the Art (SOA) method to provide cleaning and sanitizing solutions for crewmembers in-flight relies on chemicals that are regularly resupplied from Earth. This method cannot be used for extended exploration missions beyond Low Earth Orbit (LEO) because of the logistics required to maintain adequate supply of cleaning solutions. Therefore, NASA is pursuing alternative methods of cleaning solutions. While there are a variety of different methods of generating hydrogen peroxide, one specific family of technologies uses electrochemical process to oxidize clean water into hydrogen peroxide. The most promising method is to generate aqueous hydrogen peroxide in-situ, using water and oxygen, which are resources already available, and the process generates no waste: when the peroxide chemically breaks down, it releases water vapor and oxygen back into the cabin. NASA's need is to generate 3% by weight aqueous hydrogen peroxide at a rate of 1 liter per day. NASA intends to procure a system for on-site generation so that NASA can evaluate the technology to determine its feasibility for a flight system, either as a payload on-board ISS, or for future exploration mission destinations such as Lunar Habitat, Mars Transit, or Mars Habitat. Upon receipt of the system, NASA will install the system into a test facility at the Johnson Space Center, then coordinate a training session with vendor.

Deliverables
The vendor will provide the following two items:

1. A hydrogen peroxide generation system that utilizes water and air or oxygen as feed streams, which are reacted to form an aqueous solution of 3% by weight hydrogen peroxide balance water, at a rate of 1 liter per day.
2. Commissioning, Training, and Discussions by vendor engineers on-site.

Schedule
NASA to receive Deliverable 1 no later than May 31, 2025. Upon receipt of the equipment, Deliverable 2 – commissioning, training, and discussions – will be coordinated to occur in the subsequent weeks.